VendorSentinel

AI-Powered Third-Party Vendor Risk Intelligence Fabric

■■ Track 3: Web Data & Compliance | Bright Data Hackathon 2026 Created & Deployed by: Hafiz Abdul Rehman
Live UI: https://vendor-sentinel.vercel.app/ • Backend API: https://vendorsentinel.onrender.com

The Third-Party Vendor Risk Crisis
Modern enterprises lose millions annually due to blind spots in vendor cybersecurity.

$4.5M
Average Annual Cost
of a Third-Party Data Breach

■ Critical Risk Factors
SolarWinds credentials 'solarwinds123' lay exposed
on GitHub for 13 months before exploitation.
Snowflake's pre-breach client credentials circulated for weeks on hacker forums. Silent warnings are always public beforehand, but standard teams lack tools to reliably watch them.

1. Outdated PDF Questionnaires Tradition
complete

2. The Anti-Bot Defensive Wall Pastebin
trust blog

3. Factual Hallucinations in Standard AI Generic
Enterpris

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■ Track 3: Web Data & Compliance Alignment
Purpose-built compliance: how VendorSentinel harnesses Bright Data to satisfy security requirements.

Track 3 Focus Area Bright Data Integrated Technology VendorSentinel Implementation Purpose

Public Media & News Scrapes Bright Data SERP API

Queries Google search engine recursively for vendor data breaches, ransomware warnings, regulatory compliance fines, and class action lawsuits.

Blocked/CAPTCHA Paste Sites Bright Data Web Unlocker Proxy

Rotates residential proxies and automatically resolves Pastebin bot checks, scanning for leaked passwords, API keys, and vendor configuration strings.

Hiring Board Rate Limits Bright Data Web Scraper API

Ingests recent job board postings for security engineer, incident responder, and CISO positions, tracking hiring spikes that signal internal
crisis.

JS-Heavy Morale Pages Bright Data Scraping Browser

Initiates remote Playwright Chromium CDP sessions to load corporate
trust/posture pages, verifying security attestations and patch updates.

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The Autonomous Risk Intelligence Fabric
A multi-layered threat intelligence pipeline that scans continuously and reasons dynamically.

■ 1. Portfolio Onboarding Hub

Security managers add vendors simply by domain and category. The SQLite WAL
database registers the profiles instantly and schedules automated scan queues.

■ 2. Exposed Secrets Hunting

Web Unlocker proxy routes target searches to discover exposed vendor passwords or token variables circulating on public sharing repositories and code dumps.

■ 3. Talent Recruitment Spikes

Web Scraper API logs LinkedIn vacancy counts. Spike indicators (e.g. 400% surge in incident response recruitment) flag internal security overhaul periods.

■ 4. AI Mitigation Playbooks

AI scoring synthesizes all 4 vectors to grade vendor posture (1-10) and auto-builds step-by-step plain-English playbooks for corporate cybersecurity guards.

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Autonomous Threat Scan Pipeline
System-wide visual data flow, showcasing concurrent collections and verification gates.

1. Ingestion 2. Parallel Scrape 3. AI Synthesis 4. Security Alert
Admin adds vendor domain,
category, and data sensitivity (e.g.
Okta = Critical). SQLite stores and
schedules scan queue.

FastAPI concurrently spawns 4
crawling tasks via asyncio.gather
using Bright Data SERP, Unlocker,
Scraper, and Browser APIs.

All gathered evidence models feed
into GPT-4o-mini to calculate exact
risk scores (1-10), reasons, and
action items.

If computed risk score ≥ 7.0 threshold,
an automated critical warning alert is
broadcasted on the admin UI stream.

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The Coordinated Scraper Specialists
Four prompt-engineered and proxy-configured intelligence gatherers mapped to specific threat vectors.

SERP API News Specialist Queries major search engines dynamically for keywords like `data breach`, `GDPR violation`, and `cyberattack` combined with

vendor name. Standardizes unstructured news headlines into clean evidence indices.

Web Unlocker Paste Collector Accesses restricted paste boards and public GitHub code commit endpoints. Resolves CAPTCHAs and blocks automatically,

tracking leaked credentials and database secrets related to the vendor's domain.

LinkedIn Job Board Analyst Scrapes active company job listings through the Bright Data dataset API. Tracks recruitment baseline and identifies sudden

spikes in CISO, SOC analyst, and incident response vacancies.

Scraping Browser Posture
Auditor

Launches headless Playwright Chromium CDP sessions to load complex JavaScript-heavy trust and status blogs, verifying
certification validations and recently published patch advisories.

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Under-the-Hood Technical Modules

Highly optimized, production-ready modules driving data persistence and crawler stability.

FastAPI Core

Serves secure dynamic JSON API endpoints using an
asynchronous event model, providing lightning-fast
server routing under production Uvicorn containers.

■ SQLite WAL Database

Leverages `aiosqlite` WAL (Write-Ahead Logging)
transactions for safe, concurrent database read/write queries. Maintains stable local tables for `vendors`, `scan_results`, and `risk_scores` without heavy
database instances.

1. Windows Console Encoding Fix Standard
`sys.stdo

2. Windows Proactor Subprocess Integration Playwrig
We dyna
automat

3. Robust Background Task wrapping Scans tr
`scan_si
logs.

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The Cybersecurity Control Interface
A premium Bookish design system featuring modular components and high-tech threat radars.

1. Live Threat Risk Dashboard Presents overall portfolio count, unread alerts, and average risk rating. Renders custom-styled card grids highlighting critical,

warning, and low threat vendors.

2. Automated Threat Scan Radar Features an infinitely spinning golden radar sweeper, concentric active wave rings, and pulsing threat blips built entirely using

hardware-accelerated CSS keyframes.

3. Bright Data Nodes Status Integrates live status nodes with pulsing green indicator dots (`pulse-node-dot`) to prove API credentials and network

connectivity directly to judges.

4. Dynamic Playbooks Panel Drills down into detailed vendor profiles to display AI-generated threat reasoning, evidence sources, and structured playbooks

for immediate mitigation.

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Deep Bright Data & AI/ML API Integration
Normalizing global web intelligence and AI model reasoning into a single audit lifecycle.

System Module Technology Used Platform Operational Role

News Aggregator Bright Data SERP API

Crawls real-time Google results for data leaks, GDPR violations, and class action lawsuits referencing vendor names.

Secrets Hunter Bright Data Web Unlocker

Bypasses bot protections on Pastebin and GitHub to search for exposed passwords, configuration variables, and API keys.

Talent Tracker Bright Data Web Scraper API

Utilizes standard dataset scrapers to scan LinkedIn company pages, counting security, CISO, and incident response vacancies.

Posture Crawler Bright Data Scraping Browser

Launches chromium browser CDP over websocket WSS connections to parse
compliance blogs and check security patches.

AI Risk Analyst AI/ML API (GPT-4o-mini)

Receives aggregated signal logs and outputs structured threat ratings, reasoning logs, and recommended actions playbooks.

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Strategic Business Value & Market Scope
Quantifying the commercial footprint and core enterprise monetization pathways.

$4.5M
Average Cost of Breach

■ Core Verticals

• Finance & Enterprise Banking
• Health & EMR Providers
• SaaS & Cloud Infrastructure
• M&A; Due Diligence Auditing

High-Impact Operational Savings • **Labo

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Monetization & Commercial Channels • **Tiere volume,
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enterpris
• **Audit
ready fo

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Originality & Unique Selling Proposition
Comparing VendorSentinel with static database scorecards and manual PDF workflows.

System Capability Annual PDFs Legacy Scorecards VendorSentinel Fabric
Real-Time Monitoring ■ None, once a year only

■■ Delayed, relies on database
index

■ Yes, continuous 6-hour sweeps

Exposed Credentials ■ None, cannot self-report ■ Missing paste board scans ■ Yes, live Web Unlocker scans
Anti-Bot Bypassing ■ None ■ Blocks on secure boards ■ Yes, full Bright Data stack
Mitigation Playbooks ■ None ■■ Generic checklist only ■ Yes, custom AI playbooks
Hiring Distress Tracking ■ None ■ None ■ Yes, Scraper API integration

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Strategic Execution Roadmap
A step-by-step scaling plan from current production MVP to enterprise-wide integrations.

Phase 1: Production MVP Full integration of 4 Bright Data collectors + AI scoring. Complete web UI dashboards, local WAL SQLite database layer, and

public code repository push.

Phase 2: Enterprise SSO & Auth Implementing SAML/SSO multi-tenant user authentication, database security policies, granular role scopes, and custom alert

notification configurations.

Phase 3: Database Connectors Developing native enterprise connectors to link VendorSentinel to commercial data platforms like Snowflake, PostgreSQL,

BigQuery, and Salesforce APIs.

Phase 4: AI Governance Integrating dedicated AI agent security firewalls (e.g. Lobster Trap middleware) for complete cryptographic logging of prompts

and audit compliance trails.

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The Complete Technical Stack
Robust, enterprise-ready open source frameworks and cloud integration layers.

■ Visual Frontend • **Framework**: React 19 component architecture, built with lightning-fast Vite.
• **CSS Engine**: Custom vanilla CSS library coded for a premium 'Bookish' dark
theme.
• **Icons System**: Lucide React high-resolution vector graphics library.

■ Asynchronous Backend • **Language**: Python 3.11 for core intelligence workflows.
• **Web Server**: FastAPI async server framework served via Uvicorn containers.
• **Database**: SQLite with aiosqlite for safe, WAL-configured local transactional
records.

■ Scraping & AI Pipeline • **Collectors**: SERP API (Google), Web Unlocker (Pastebin), Scraper API (LinkedIn),
and Scraping Browser (Playwright CDP).
• **AI Core**: AI/ML API (GPT-4o-mini) for automated risk scoring and playbook
synthesis.

■ Deployment & Lifecycle • **Frontend Hosting**: Vercel cloud network (linked to GitHub commits auto-rebuild).
• **Backend Deployment**: Render cloud services running stable virtual environments.
• **Secret Protection**: Dotenv environment parameters (.env.example).

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Interactive Demonstration Guide

Step-by-step walkthrough to test every core platform module live.

Step 1: Open Vercel App Open your live frontend link https://vendor-sentinel.vercel.app. Click **`Enter Security

Dashboard →`** to bypass the welcome gate.

Step 2: Onboard Figma (Low
Risk)

Go to `/vendors`, click **`+ Add Vendor`**, and onboard `Figma, Inc.` (domain: `figma.com`) with `Low` sensitivity. Scanners
verify clean SOC 2 compliance, scoring **2.0 Low Risk**.

Step 3: Onboard Auth0 (Critical) Onboard `Auth0 by Okta` (domain: `auth0.com`) with `Critical` sensitivity. Key leak queries trigger warnings, and the AI Engine

generates a **8.0 Critical Risk** rating.

Step 4: Audit Scan Center Navigate to `/scan`. Verify that the golden spinning radar is fully operational, click **`Trigger System-Wide Scan`**, and watch

background audits poll live to completed states.

Step 5: Verify Node Health Go to `/settings`. Confirm the golden threshold slider operates correctly, and see all **Bright Data Nodes** marked with pulsing

green active indicators.

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VendorSentinel: Security with Certainty

Redefining how enterprises validate, reconcile, and monitor third-party cybersecurity postures.

■ Always-On Sweeping
Continuous 6-hour sweeps across paste boards,
search engines, compliance blogs, and talent listings instead of stale PDF checklists.

■■ Deep Bright Data Integration
Defeats bot blocks and CAPTCHAs natively using
SERP API, Web Unlocker, Web Scraper, and headless Headless Scraping Browser CDP.

■ Evidence-Backed AI Playbooks
Generates clean risk grades (1.0-10.0), detailed
structural justifications, and custom mitigation
playbooks under a robust offline rule-based fallback
framework.

“VendorSentinel doesn't just scan vendors — it secures the supply chain.”

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